

Kinematics

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Kinematics

Studying the **motion** of an **object** in space time without looking after the causes: Laws governing geometrical motion aspects.

Example: In Formula 1, to identify the winner, we look at the front wing as it crosses the finish line. However, in kinematics, we consider,

Object: is assumed as a point-like regardless of its size: **a particle having infinitesimal size.**

Motion: When two parallel trains move in the same direction and two passengers from each train are standing in front of one another, there appears to be no relative motion between them. However, from the perspective of someone at the train station, both trains are moving away.

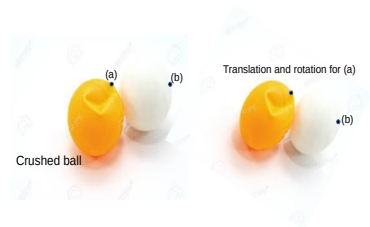
Motion

Changing of positions in a specific time for an external observer: frame of reference.

Rigid body

If we crush a ball and make it moving forward

- All parts of the ball change their positions.
- Any point (a) is subject to a combined motion of translation and rotation, different from the non-crushed ball (white).



When the shape of an object doesn't change it is called **rigid-body**

In this case all points with the same radius have similar circular motions.
The balance point of an object is referred in physics to **the center of mass**.

The center-of-mass motion

of any rigid body can be split into rotation and center-of-mass motion.

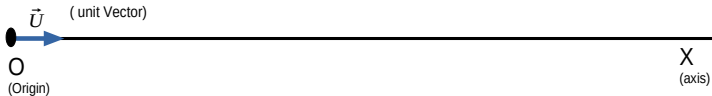
Frames of reference

To study any motion we need:

- an origin O .
- a coordinate system of orthogonal base like Cartesian system (Ox, Oy, Oz).
- unit vectors ($\vec{i}, \vec{j}, \vec{k}$)

A one dimensional motion is characterized only by one coordinate.

Example



This is a **straight line** motion. We need two variables: the position x relative to the origin, and t which measures a point in time.

Position Vector

An object (M) is in a position M_o at t_o , and at t_1 is in M_1 ..., $M_o, M_1, M_2, \dots$, specify only positions,



$\overrightarrow{OM_o}$ is the vector position at t_o (similar $\overrightarrow{OM_1}$ at t_1)

If we consider the distance between O and M_o equal to x_o then,

$$\overrightarrow{OM_o} = x_o \vec{u}$$

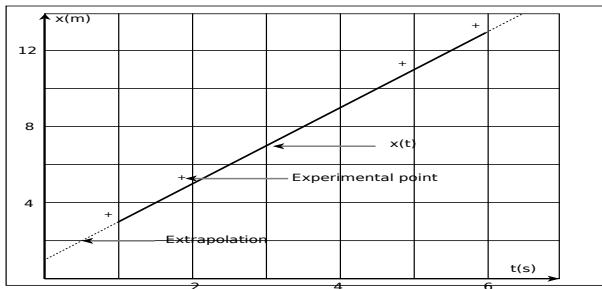
- $x_o(t_o)$ is the **coordinate** at (t_o)
- t_o is the initial time and x_o is the **initial coordinate** (M_o is the initial position).
- The set of points mobile ($M_o, M_1, M_2, \dots$) form the **mobile trajectory**.

Example

- ① A mobile coordinates are given along the x-axis by x versus t ,

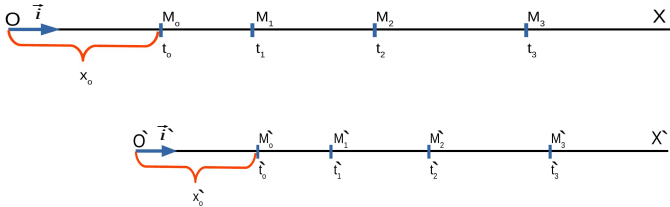
$t(s)$	1	2	5	6
$x(m)$	3	5	11	13
$\vec{OM}$	$\vec{r}(1) = 3\vec{i}$	$\vec{r}(2) = 5\vec{i}$	$\vec{r}(5) = 11\vec{i}$	$\vec{r}(6) = 13\vec{i}$

- ② **space diagram or position-time** is the plot of x versus t .
- ③ **Kinematic Equation**: corresponds to the function $x = f(t)$, i.e.
 $x(t) = 2t + 1$. and the vector position is: $\vec{r}(t) = x(t)\vec{i} = (2t + 1)\vec{i}$



Rate of displacement

Consider two mobiles, denoted as B_1 and B_2 , moving parallel to each other along straight lines. The position of B_1 is represented by M_0 at time t_0 , and at time t_1 , it is at M_1 , measured from point O . Similarly, the positions of B_2 are represented by M'_0 at time t'_0 , and at time t'_1 , it is at M'_1 , measured from point O' .



This situation is similar to what is encountered in a Formula One race. How can we compare two parallel motions if they don't start at the same time and have different origins? Calculate the **rate**

Velocity vector

The **displacement** is the change in position of a mobile in some time interval: $\Delta x = x_f - x_i$ (different from **Distance** which is the length of a path)

Average velocity

Rate of displacement variation:

$$v = \frac{x_f - x_i}{t_f - t_i} = \frac{\Delta x}{\Delta t} (m/s),$$

Average speed:

$$V = \frac{\text{total distant}}{\text{total time}}$$

which is the average velocity in the interval $[t_i, t_f]$

The average velocity vector is:

$$\vec{v} = \frac{\Delta x}{\Delta t} \vec{i}$$

- ① shows the motion direction which is given by $\vec{i}$.
- ② The sign of Δx indicates the direction of motion. (opposite to $\vec{i}$ if $\Delta x < 0$)

instantaneous velocity

If the instant t_f and t_i are so close that $\Delta t = (t_f - t_i) \rightarrow 0$, which means that we are approaching the **middle** of the interval $[t_i, t_f]$, then

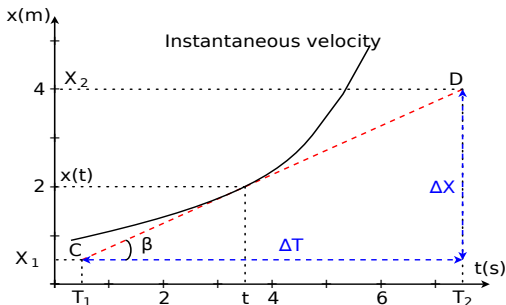
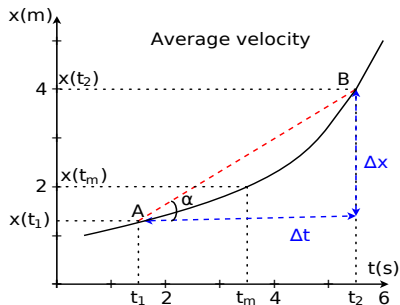
$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx(t)}{dt}$$

The instantaneous velocity is the limiting value of the ratio $\Delta x(t)/\Delta t$ as it approaches zero. It is also the first derivative of position x with respect to t .

The instantaneous velocity is:

$$\vec{v} = \frac{dx(t)}{dt} \vec{u}$$

Corresponding to the **slope (tangent)** of the space-time diagram at the point of coordinates $(t, x(t))$.



$$t_1 = 1.5s, \quad t_m = 3.5s, \quad t_2 = 5.5s, \quad x(t_1) = 1.3m, \quad x(t_m) = 2m, \quad x(t_2) = 4m.$$

$$v_m(t_1, t_2) = \frac{x(t_2) - x(t_1)}{t_2 - t_1} = \frac{\Delta x}{\Delta t} = \tan(\alpha) = \frac{4 - 1.3}{5.5 - 1.5} = 0.68m/s$$

$$T_1 = 0.5s, \quad t = 3.5s, \quad T_2 = 7.5s, \quad X_1 = 0.5m, \quad x(t) = 2m, \quad X_2 = 4m.$$

$$v(t) = \frac{X_2 - X_1}{T_2 - T_1} = \frac{\Delta X}{\Delta T} = \tan(\beta) = \frac{4 - 0.5}{7.5 - 0.5} = 0.50m/s$$

Example

A mobile is moving along the x axis. The equation of motion is given by

$$x = -4t + 2t^2(m)$$

x is in meters and t is in seconds. At the 1st second, the mobile is moving towards negative x , then makes a pause during 1 s. After that the motion continuous in the sens of positive x for $t > 1$. (A) Determine the displacement of the mobile in the time intervals 0 and 1s, 1s and 3s. For the interval $[0, 1]$,

$$\Delta x_1 = x_1 - x_0 = -4 \times 1 + 2 \times 1^2 - 0 = -2m$$

For the interval $[1, 3]$, s:

$$\Delta x_1 = x_2 - x_1 = -4 \times 3 + 2 \times 3^2 - (-4 \times 1 + 2 \times 1^2) = +8m$$

The graph: $x = f(t)$ provides also these values directly.

Acceleration Vector

The acceleration of a mobile is the change in velocity v_x divided by the time interval Δt during which that change occurred.

$$\text{average: } \bar{a}_m = \frac{\Delta v}{\Delta t}, \quad \text{instantaneous: } \vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{OM}}{dt} = \frac{d^2x}{dt^2} \vec{u} [m/s^2]$$

Second derivative of displacement with respect to time.

The acceleration of a mobile is the slope at a point of $v(t)$ graph.

Acceleration vector shows the motion type as given by the table :

(C: is a constant)

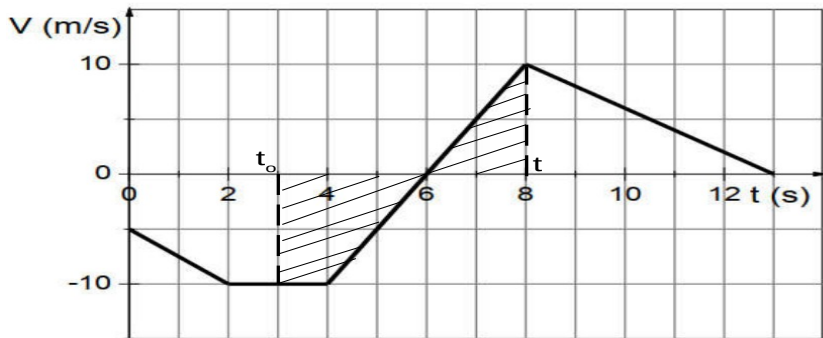
Motion	Uniform	var Uniform.	speed up	slow down	Sens +	Sens -
x	$\Delta x = C$	-	$ \Delta x \nearrow$	$ \Delta x \searrow$	$\Delta x > 0$	$\Delta x < 0$
v	$v = C$	$\Delta v = C$	$ v \nearrow$	$ v \searrow$	$v > 0$	$v < 0$
a	$a_t = 0$	$a_t = C$	$a_t v > 0$	$a_t v < 0$	-	-

Inverse Formula

If the initial position is known $x_0 = x(t_0)$ at a given time t_0 , then we can find the x position at any time by using the $v(t)$ graph $x(t)$:

$$x(t) = \int v(t) dt \iff x(t) - x(t_0) = \int_{t_0}^t v(t) dt = A(t_0, t, v)$$

$A(t_0, t, v)$ is the hatched surface: area under the curve, definite integral



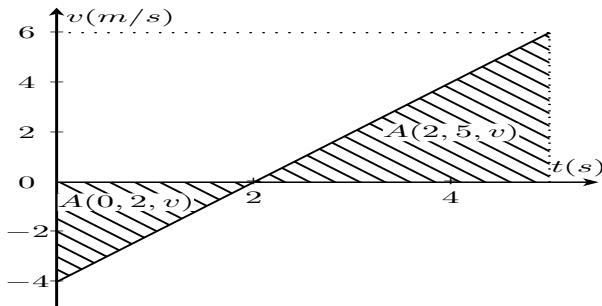
Example

Let $t_0 = 2s$, $x(2) = 3m$ et $v(t) = 2t - 4$. Plot $v(t)$ in the time interval between 0 and 5s. Find $x(0)$ and $x(5)$ by surface calculation. Give the function $x(t)$.

Example

Let $t_0 = 2s$, $x(2) = 3m$ et $v(t) = 2t - 4$. Plot $v(t)$ in the time interval between 0 and 5s. Find $x(0)$ and $x(5)$ by surface calculation. Give the function $x(t)$.

1. Velocity graph



$$2. \quad x(2) - x(0) = A(0, 2, v) = -\frac{4 \times 2}{2} \implies x(0) = 7m ,$$

$$x(5) - x(2) = A(2, 5, v) = +\frac{6 \times 3}{2} \implies x(5) = 12m.$$

Example ...

$$3. \quad x(t) - x(2) = \int_2^t (2t - 4) dt = [t^2 - 4t]_2^t = t^2 - 4t + 4.$$

$x(t) = t^2 - 4t + 7$, so one finds $x(0) = 7m$ et $x(5) = 12m$.

Other approach: $x(t) = \int (2t - 4) dt = t^2 - 4t + C$ with

$$x(2) = 3 \implies -4 + C = 3 \implies C = 7, \text{ then } x(t) = t^2 - 4t + 7.$$

From v function and given the acceleration a and $v(t_1)$

In the interval $[t_1, t]$

$$dv(t) = a(t)dt \iff v(t) - v(t_1) = \int_{t_1}^t a(t)dt = A(t_1, t, a)$$

$A(t_1, t, v)$ is the surface limited by x-axis (time) and the curve $a(t)$.

Vertical lines t_1 et t correspond to the boundaries. To find $x(t)$ from $a(t)$, we make the integral of $v(t)$. For that, we need both $x(t_0)$ and $v(t_1)$; two initial conditions.

Kinematic Equations Derived from Calculus

$$a = \frac{dv}{dt} \Rightarrow a = \frac{dv}{dx} \frac{dx}{dt} \Rightarrow a = \frac{dv}{dx} v \Rightarrow a dx = v dv$$

By integration having, $x(t_0) = x_0$, $v(t_0) = v_0$, $x(t_1) = x_1$ and $v(t_1) = v_1$ one finds

$$\int_{v_0}^{v_1} v dv = \int_{x_0}^{x_1} a dx \Rightarrow \frac{1}{2} v_1^2 - \frac{1}{2} v_0^2 = \int_{x_0}^{x_1} a dx$$

For a constant acceleration, we obtain:

$$v_1^2 - v_0^2 = 2a(x_1 - x_0)$$

Constant velocity

By integration,

$$x(t) - x(t_0) = \int_{t_0}^t v dt = v \int_{t_0}^t dt = v \times (t - t_0)$$

For example $v = 3 \text{ m/s}$, $x_0 = 3$ at $t_0 = 1$ then,

$$x(t) - 3 = \int_1^t 3 dt = 3 \int_1^t dt = 3 \times (t - 1)$$

to find,

$$x(t) = 3 \times$$

In general, the position of the mobile at time t corresponds to.

$$x(t) = v \times (t - t_0) + x(t_0)$$

Motion with Constant Acceleration

For a constant acceleration:

$$a = \frac{v(t) - v(t_0)}{t - t_0}, \quad \text{put } v(t_0) = t_0 \text{ and } t_0 = 0$$

then, $v(t) = a t + v_0$, a is the slop of the velocity-time graph. as $v(t) = dx/dt$ is function of time, by integration one finds

$$x(t) - x(t_0) = \int_{t_0}^t v dt = \int_{t_0}^t (a \times (t - t_0) + v(t_0)) dt$$

For $a = 5 \text{ m/s}$, $v(2) = 1 \text{ m/s}$ and $x(2) = 3 \text{ m/s}$

$$x(t) - 3 = \int_2^t v dt = \int_2^t (5 \times (t - 2) + 3) dt$$

Let $u(t) = t - t_0 = t - 2$ then $\frac{du}{dt} = 1 \Rightarrow du = dt$, to get,

$$x(t) - 3 = \int_{u(t_0)=0}^{u(t)=t-2} (5 \times u + 3) du = \left[\frac{5}{2} u^2 + 3u \right]_0^{t-2} =$$

$$x(t) = \frac{5}{2}(t - 2)^2 + 3(t - 2) + 3$$

Freely Falling Objects

All objects dropped near the Earth's surface fall with the same constant acceleration

From kinematics equation: $x(t) = \frac{a}{2}(t - t_0)^2 + v(t_0)(t - t_0) + x(t_0)$

A freely falling object moves under the influence of gravity alone.

We put $a = g$ (or $-g$) (depending on the axis direction choice)

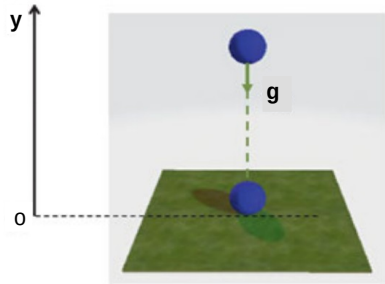
Example

For a ball dropped from shoulder height: $g = 10 \text{ m/s}^2$, $v(0) = 0$ and $y(0) = 1.5 \text{ m}$.

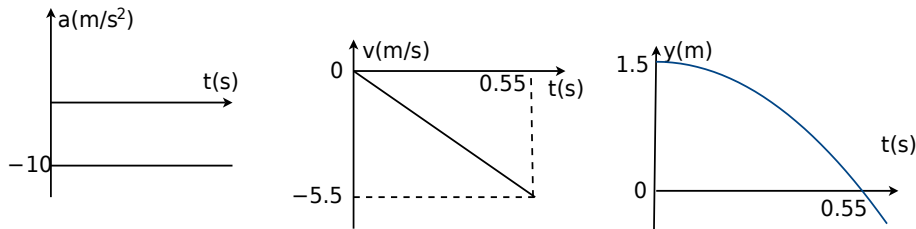
$$v(t) - v(0) = \int_0^t a dt \implies v(t) = -10t \text{ and } y(t) - y(0) = \int_0^t v dt \implies y(t) = -5t^2 + 1.5.$$

The time for the ball to reach the ground is: $y(t) = 0$,

$$-5t^2 + 1.5 = 0 \text{ and } t = \sqrt{1.5/5} \text{ s}$$



The graphs corresponding to the free fall motion are,



- Acceleration : constant $a = -9.8(\sim 10) \text{ m/s}^2 \Rightarrow$ horizontal line.
- Velocity: linear with g as slope.
- Position-time graph : v is linear, $y(t)$ is a parabola with $y(0) = 1.5 \text{ m}$.

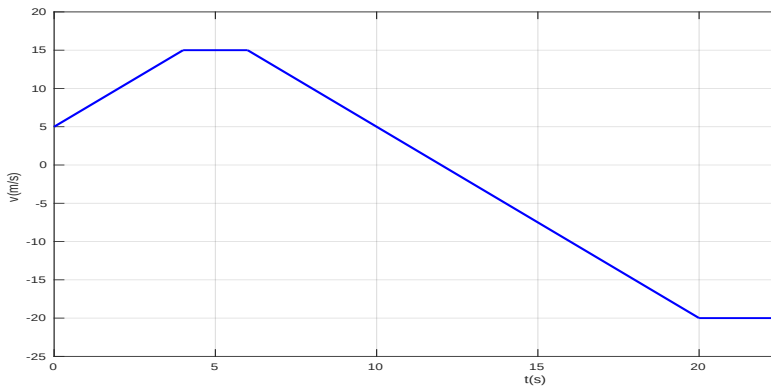
Summary

$t(s)$	< 0.55	0.55	> 0.55
$v(t)$	$+$	0	$?$
$z(t)$	$\searrow$	$y(0.55) = 0 \text{ m}$	$?$

A velocity-time graph of a mobile (A) of straight-line motion is given by the figure below.

- 1 Plot the acceleration-time graph.
- 2 What are the phase motion.
- 3 Find the mobile positions at $t = 6 \text{ s}$, $t = 10 \text{ s}$ and $t = 20 \text{ s}$. We consider that at $t = 0 \text{ s}$ we have $x_{A0} = 10 \text{ m}$.
- 4 At what time the mobile turn back?
- 5 When did the mobile cross the origin?
- 6 Show on the mobile path, position, velocity and acceleration vectors at $t = 10 \text{ s}$. Choose: $1 \text{ cm} \rightarrow 20 \text{ m}$, $1 \text{ cm} \rightarrow 2 \text{ m/s}$, and : $1 \text{ cm} \rightarrow 1 \text{ m/s}^2$.
- 7 At $t = 0 \text{ s}$, a second mobile is at $x_{B0} = 0 \text{ m}$ having a velocity $v_B = 10 \text{ m/s}$, constant during the motion along x -axis. Find the velocity of (A) in a frame moving with (B) ($\vec{V}_{A/B}$) at $t = 20 \text{ s}$?

Velocity-time graph



The velocity-time Graph

1. Acceleration-time graph: $a=f(t)$.

when $t \in [0, 4]$, $a_1 = \Delta v / \Delta t = (15 - 5) / (4 - 0) = 5/2 \text{ m/s}$

when $t \in [4, 6]$, and $t \in [20, 24]$, $a_2 = a_4 = 0 \text{ m/s}$

when $t \in [6, 20]$, $a_3 = \Delta v / \Delta t = (0 - 15) / (12 - 6) = -5/2 \text{ m/s}$

2. Motion phases:

$t \in [0, 4]$, straight-line accelerated motion
as $a_1 = \text{constant}$ and $\vec{a} \cdot \vec{v} > 0$

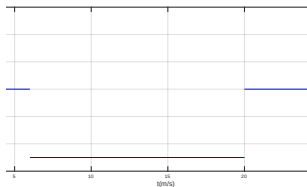
$t \in [4, 6]$, and $t \in [20, 26]$, $a_2 = a_4 = 0 \text{ m/s}$ uniform motion.

$t \in [6, 12]$, Mobile slowing down

Because $a_3 = \text{constante}$ and $\vec{a} \cdot \vec{v} < 0$

$t \in [12, 20]$, Accelerated motion

because $a_3 = \text{constant}$ and $\vec{a} \cdot \vec{v} > 0$.



Solution 2

3. Mobile: as

$$v = \frac{dx}{dt} \quad \text{then,} \quad \Delta x = \int_{t_1}^{t_2} v dt = \text{limited surface by the curve and } [t_1, t_2]$$

Remember that at $t_o = 0$ the mobile was at $x_{Ao} = 10 \text{ m}$.

Therefore, at $t = 6 \text{ s}$:

$$x_1 = v * \Delta t + x_{Ao} = 4 * 5 + (15 - 5) * 4/2 + (6 - 4) * 15 + 10 = 80 \text{ m}.$$

$$t = 10 \text{ s}, x_2 = x_1 + 5 * (10 - 6) + (15 - 5)(10 - 6)/2 = 120 \text{ m}$$

$$t = 20 \text{ s}, x_3 = x_2 + 5 * (12 - 10)/2 - 20 * (20 - 12)/2 = 45 \text{ m}$$

4. When $v = 0$ the mobile turn back i.e; at $t = 12 \text{ s}$

5. Crossing the origin: The distance is equal to that traveled at $t = 12 \text{ s}$.

From the graph

$$\Delta x = (15 - 5)4/2 + 5 * 4 + (6 - 4)15 + 15(12 - 6)/2 + 10 = 115. \text{ and}$$

$$\Delta x = |-20(t - 12)| \quad t = \Delta x / 20 + 12 = 115/20 + 12 = 17.75 \text{ s}$$

6. Mobile velocity: $x_A = x_B + x_{A/B}$ by derivation: $v_A = v_B + v_{A/B}$. So,
 $v_{A/B} = V_A - V_B = -20 - 10 = -30 \text{ m/s}.$

As the motion is in a plan, two coordinates are required for the study.

Example

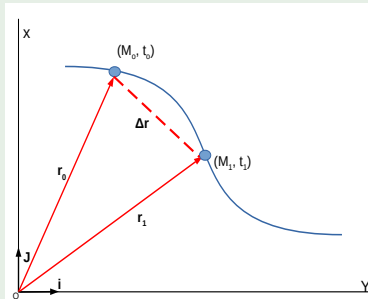
A mobile is moving alongside the path of the figure.

$M_o(M_1)$ is the position at $t_o(t_1)$

The vector position is given $\vec{r}$. At a given time t_1 we have

$$\vec{r}_1 = x_1 \vec{i} + y_1 \vec{j}$$

x_1, y_1 correspond to the mobile coordinates in the frame (xoy) at t_1 .



Displacement, velocity, acceleration

In the interval $[t_0, t_1]$ the displacement vector is

$$\Delta \vec{r} = \vec{r}_1 - \vec{r}_0$$

and the average velocity vector corresponds to,

$$\vec{v}_m(t_1, t_0) = \frac{\vec{r}(t_1) - \vec{r}(t_0)}{t_1 - t_0} = \frac{\Delta \vec{r}}{\Delta t}$$

When $\Delta t \rightarrow 0$ we have the instantaneous velocity t

$$\vec{v}(t) = \lim_{\Delta t \rightarrow 0} \vec{v}_m(t, t + \Delta t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

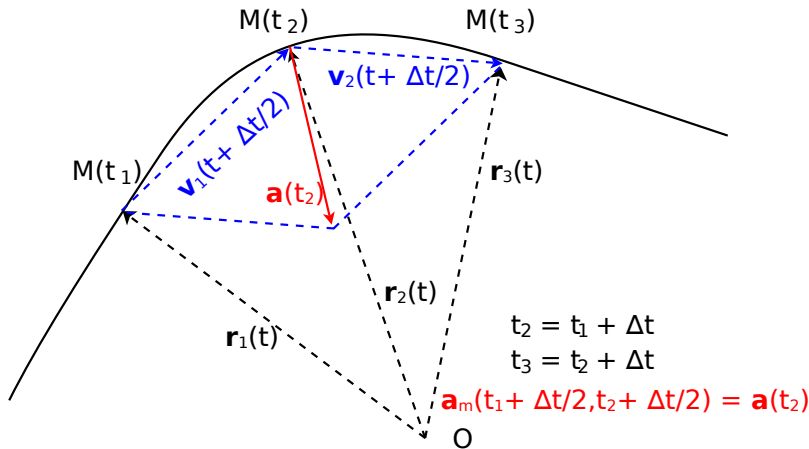
Which correspond to the trajectory slope.

The acceleration vector is

$$\text{average: } \vec{a}_m(t_1, t_2) = \frac{\vec{v}(t_2) - \vec{v}(t_1)}{t_2 - t_1}$$

$$\text{Instantaneous: } \vec{a}(t) = \lim_{\Delta t \rightarrow 0} \vec{a}_m(t, t + \Delta t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

The mobile positions at t_1 , t_2 et t_3 correspond to M_1 , M_2 , M_3 , respectively (figure). One obtains the velocity vectors.



By projection we can study the motion independently in each axis ox or oy : straight-line motion $x'Ox$ and $y'Oy$. In the frame (oxy) the vectors:

position	velocity	acceleration
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$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}, \quad \vec{v}(t) = v_x(t)\vec{i} + v_y(t)\vec{j}, \quad \vec{a}(t) = a_x(t)\vec{i} + a_y(t)\vec{j}$$

with x is the **abscissa** and y is the est **ordinate**: Cartesian coordinates.
motion equations :

$$x = x(t), \quad y = y(t)$$

Parametric equations.

By eliminating the time, we get the equation of trajectory $y = f(x)$ (or $x = g(y)$).

As $\frac{d\vec{i}}{dt} = \frac{d\vec{j}}{dt} = \vec{0}$, and $\vec{v} = \frac{d\vec{r}}{dt}$ and $\vec{a} = \frac{d\vec{v}}{dt}$, etc..., then:

- According to x-axis

$$v_{xm}(t_1, t_2) = \frac{x_2 - x_1}{t_2 - t_1}, v_x(t) = \frac{dx(t)}{dt}, x(t_2) - x(t_1) = \int_{t_1}^{t_2} v_x(t) dt$$

$$a_{xm}(t_1, t_2) = \frac{v_{x2} - v_{x1}}{t_2 - t_1}, a_x(t) = \frac{dv_x(t)}{dt}, v_x(t_2) - v_x(t_1) = \int_{t_1}^{t_2} a_x dt$$

- According to y-axis

$$v_{ym}(t_1, t_2) = \frac{y_2 - y_1}{t_2 - t_1}, v_y(t) = \frac{dy(t)}{dt}, y(t_2) - y(t_1) = \int_{t_1}^{t_2} v_y(t) dt$$

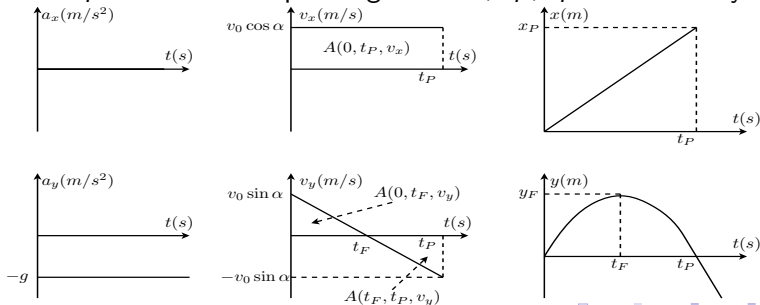
$$a_{ym}(t_1, t_2) = \frac{v_{y2} - v_{y1}}{t_2 - t_1}, a_y = \frac{dv_y}{dt}, v_y(t_2) - v_y(t_1) = \int_{t_1}^{t_2} a_y(t) dt$$

Example: projectile motion

Let : $(x(0) = 0, y(0) = 0)$, $(v_x(0) = v_0 \cos \alpha, v_y(0) = v_0 \sin \alpha)$,
 $(a_x(t) = 0, a_y(t) = -g)$. Analytical method : $x(t) = (v_0 \cos \alpha)t$ et
 $y(t) = -\frac{g}{2}t^2 + (v_0 \sin \alpha)t$. The trajectory: from $t = x/(v_0 \cos \alpha)$ one
 obtains

$$y = \frac{-gx^2}{2v_0^2 \cos^2 \alpha} + \tan \alpha x; \quad 0 \leq \alpha \leq \frac{\pi}{2}$$

By using the graph **the maximum height** is $h = y_F$ and **the horizontal range** is $D = x_P$ and the corresponding instants, t_F , t_P are shown by



figures:

Example (2)

The maximum vertical displacement : y_F and t_F correspond to $v_y = 0$. v_y graph gives t_F

$$-g = \frac{0 - v_0 \sin \alpha}{t_F - 0} \implies t_F = \frac{v_0 \sin \alpha}{g}$$

By using surface calculation we get

$$y_F - y_0 = A(0, t_F, v_y) = \frac{(v_0 \sin \alpha) \times t_F}{2} \implies h = y_F = \frac{(v_0 \sin \alpha)^2}{2g}$$

The range: x_P and t_P correspondent to $y_P = 0$. Therefore,
 $y_F - y_0 = y_F - y_P$, c.-a.-d $A(t_0, t_F, v_y) = -A(t_F, t_P, v_y)$. So,

$$t_P = 2t_F = \frac{2v_0 \sin \alpha}{g}$$

The graph v_x yields,

$$x_P - x_0 = A(0, t_P, v_x) = v_0 \cos \alpha \frac{2v_0 \sin \alpha}{g}$$

Exercise

and using $2 \cos \alpha \sin \alpha = \sin(2\alpha)$, one finds

$$D = x_P = \frac{v_0^2 \sin(2\alpha)}{g}$$

For the same velocity v_0 , the maximum range corresponds to

$$\sin 2\alpha = 1 \implies \alpha = \frac{\pi}{4} = 45^\circ; \text{ and } D_{\max} = \frac{v_0^2}{g}$$

Exercise

A sprinter makes a jump, having an angle to horizontal line of $\alpha = 20^\circ$ and a velocity $v_o = 11 \text{ m/s}$.

- ① Find the maximum height?
- ② Find the score?

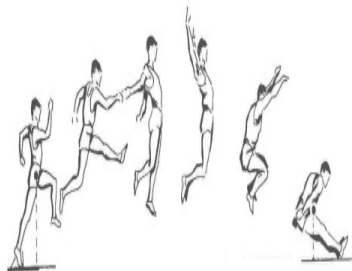
This motion is similar to a projectile motion.

Parallel to horizontal line (ox): $x = \cos(\alpha)t$

parallel to the vertical line (oy): the velocity is $v_y = -gt + v.\sin(\alpha)$ and $y = -gt^2/2 + v.\sin(\alpha)t$

1) At the maximum height $v_y = 0$ which occurs at $t_s = v.\sin(\alpha)/g = 11.\sin(20^\circ)/g = 0.384 \text{ s}$. We use the motion equation: y to obtain $h = v_y(t = 0.384) = -9.8 * (0.384)^2/2 + 11 * \sin 20^\circ * 0.384 = 0.722 \text{ m}$

2) The score corresponds to the horizontal distance when the sprinter feet reach the land: the range. As the maximum height is a symmetrical point then $t_p = 2t_s$ and $x_p = \cos(20^\circ) * (2 * 0.384) = 7.94 \text{ m}$



Circular Motion

The path is a circle of equation

$$x^2 + y^2 = R^2 \quad R: \text{radius}$$

In Cartesian coordinates (by r projection)

$$x = R \cos(\theta(t)), \quad y = R \sin(\theta(t))$$

$\theta = \theta(t)$ (rad).

Trajectory:

$$x^2 + y^2 = R^2(\cos^2 \theta + \sin^2 \theta) = R^2$$

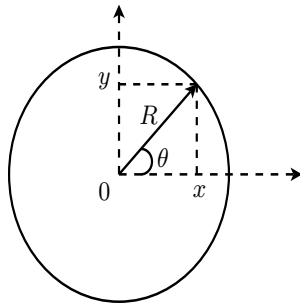
Velocity

$$v_x = \dot{x} = -\dot{\theta}R \sin \theta, \quad v_y = \dot{y} = \dot{\theta}R \cos \theta$$

where

$$v = \pm \sqrt{v_x^2 + v_y^2} = \dot{\theta}R$$

$\dot{\theta}(\text{rad.s}^{-1})$ is the angular velocity.



For $\dot{\theta} > 0$, (opposite de clockwise) then acceleration corresponds to:

$$a_x = -\ddot{\theta}R \sin \theta - \dot{\theta}^2 R \cos \theta; \quad a_y = \ddot{\theta}R \cos \theta - \dot{\theta}^2 R \sin \theta; \quad \ddot{\theta}(\text{rad.s}^{-2})$$

Exercise

- ① A mobile is moving on a circle of radius $r = 10 \text{ m}$ with a constant velocity $v = 15 \text{ m/s}$. Find the angular velocity.
- ② The bike wheel makes 8 rotations per second. If the wheel diameter is equal to 41 cm , calculate the angular speed.

Response

- ① The angular speed is : $\omega = v/r; \quad \omega = 1.5 \text{ rad/s}$
- ② For the bike's wheel:
 $\omega = N(\text{rotations}' \text{ number}).2\pi/r; \quad \omega = 16\pi/41 \text{ rad/s}$

Sinusoidal Motion

can be obtained from the projection of uniform circular motion on ox (or oy) axis:

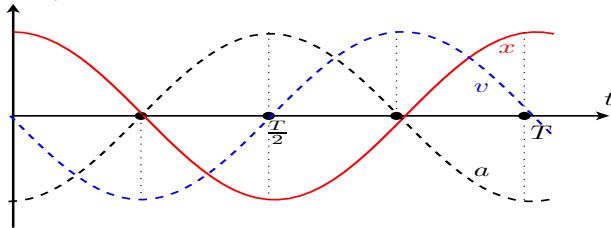
$$x(t) = x_0 \cos(\omega t + \varphi)$$

x_0 : being the magnitude, $(\omega t + \varphi)$: is the phase, φ : is the initial phase and $\omega = 2\pi\nu = \frac{2\pi}{T}$ corresponds to: pulsation and ν is the frequency: the number of periods per second.

$$\text{velocity : } v(t) = \dot{x} = -\omega x_0 \sin(\omega t + \varphi)$$

$$\text{acceleration : } a(t) = \dot{v} = -\omega^2 x_0 \cos(\omega t + \varphi) \Rightarrow \ddot{x}(t) = -\omega^2 x(t)$$

For $\varphi = 0$, the angular-time graph corresponds to:



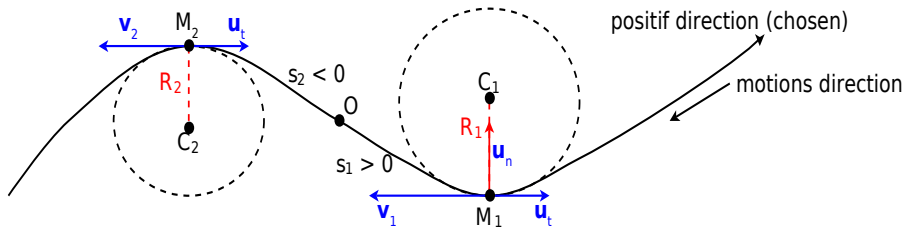
$$x : 1.5 \text{ cm} \longrightarrow x_0$$

$$v : 1.5 \text{ cm} \longrightarrow \omega x_0$$

$$a : 1.5 \text{ cm} \longrightarrow \omega^2 x_0$$

Intrinsic coordinate

associate to the mobile trajectory.



O is the origin (chosen).

The curved abscissa s corresponds to the length of $\widehat{OM}$

$$s_1(t) = \widehat{OM}_1(t)$$

the velocity is

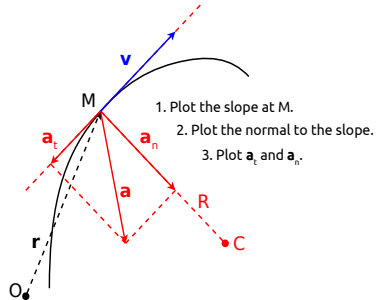
$$v = \frac{ds}{dt}$$

where $\vec{u}_t$ is a unit vector: tangent to the path at M and oriented to the positive direction (chosen).

Velocity

$$\vec{v} = v \vec{u}_t \quad \vec{u}_t = \frac{\vec{v}}{v} = \begin{cases} \frac{\vec{v}}{\|\vec{v}\|} & v > 0 \\ -\frac{\vec{v}}{\|\vec{v}\|} & v < 0 \end{cases}$$

The vector $\vec{u}_n$ is normal to $\vec{u}_t$ at M oriented to the curve center $C(t)$ of the trajectory at M .



$$\frac{d\vec{u}_t}{dt} = \frac{v}{R} \vec{u}_n, \quad \frac{d\vec{u}_n}{dt} = -\frac{v}{R} \vec{u}_t$$

R being the radius at M . **The acceleration vector** is

$$\vec{a} = \frac{dv}{dt} \vec{u}_t + v \frac{d\vec{u}_t}{dt} = \frac{dv}{dt} \vec{u}_t + \frac{v^2}{R} \vec{u}_n$$

Which can be written as

$$\vec{a} = a_t \vec{u}_t + a_n \vec{u}_n, \quad a_t = \frac{dv}{dt} = \frac{d^2 s}{dt^2}, \quad a_n = \frac{v^2}{R}$$

Properties

- The tangential acceleration a_t shows the rate of change of v (magnitude),
- The normal acceleration a_n shows the trajectory type ($\vec{v}$ direction).
- We summarize by:

straight-line	Curved	Circular
line	curved	circular
Direction of $\vec{v}$ Cte	Direction of $\vec{v}$ changing	Direction of $\vec{v}$ chagning
$a_n = 0$	$a_n \neq 0$	$a_n \neq 0$ and $R = C$

Motion	Uniform	varying unif.	Varying	Accelerated	slow-down
Justification	$a_t = 0$	$a_t = C$	$a_t \neq 0$	$a_t v > 0$	$a_t v < 0$

We note, $a^2 = a_t^2 + a_n^2$, $a_n = \frac{v^2}{R}$ et $a_t = \vec{a} \cdot \frac{\vec{v}}{v} = \vec{a} \cdot \vec{u}_t$

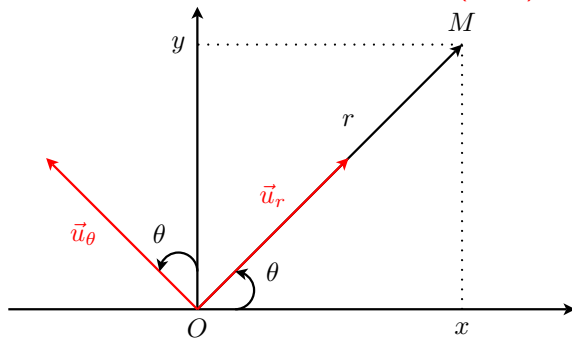
Polar coordinates

Corresponds to (r, θ) where $r = \|\vec{OM}\| \in [0, \infty[$ et $\theta = (\vec{i}, \vec{OM}) \in [0, 2\pi[$.

Unit Vectors:

radial $\vec{u}_r$ in the same direction as $\vec{r}$.

transversal $\vec{u}_\theta$ is perpendicular to $\vec{u}_r$ having the same orientation as the conventional angle positive direction ($\perp \vec{u}_r$).



From polar to Cartesian

$$\begin{aligned}x &= r \cos \theta, & y &= r \sin \theta \\ \vec{u}_r &= \cos \theta \vec{i} + \sin \theta \vec{j} \\ \vec{u}_\theta &= -\sin \theta \vec{i} + \cos \theta \vec{j}\end{aligned}$$

Thus,

$$\frac{d\vec{u}_r}{d\theta} = \vec{u}_\theta, \quad \frac{d\vec{u}_\theta}{d\theta} = -\vec{u}_r$$

So

- The derivative of a unit vector to the angle is similar to a rotation by $\frac{\pi}{2}$ (in the positive direction).
- Time derivatives are,

$$\frac{d\vec{u}_r}{dt} = \frac{d\theta}{dt} \frac{d\vec{u}_r}{d\theta} = \dot{\theta} \vec{u}_\theta, \quad \frac{d\vec{u}_\theta}{dt} = \frac{d\theta}{dt} \frac{d\vec{u}_\theta}{d\theta} = -\dot{\theta} \vec{u}_r$$

One can find the acceleration component.

Acceleration Vector

Position vector is

$$\vec{r} = r\vec{u}_r$$

Velocity vector is $\vec{v} = \frac{d\vec{r}}{dt} = \dot{r}\vec{u}_r + r\dot{\vec{u}}_r = \dot{r}\vec{u}_r + r\dot{\theta}\vec{u}_\theta$. or

$$\vec{v} = v_r\vec{u}_r + v_\theta\vec{u}_\theta$$

$$v_r = \frac{dr}{dt}$$

$$v_\theta = r\frac{d\theta}{dt}$$

Acceleration vector $\vec{a} = \frac{d\vec{v}}{dt} = \dot{v}_r\vec{u}_r + v_r\dot{\vec{u}}_r + \dot{v}_\theta\vec{u}_\theta + v_\theta\dot{\vec{u}}_\theta$. One finds

$$\vec{a} = a_r\vec{u}_r + a_\theta\vec{u}_\theta$$

$$a_r = \frac{dv_r}{dt} - v_\theta\frac{d\theta}{dt} = \frac{d^2r}{dt^2} - r\left(\frac{d\theta}{dt}\right)^2$$

$$a_\theta = \frac{dv_\theta}{dt} + v_r\frac{d\theta}{dt} = r\frac{d^2\theta}{dt^2} + 2\frac{dr}{dt}\frac{d\theta}{dt}$$

Exercise (from an exam, 2019)

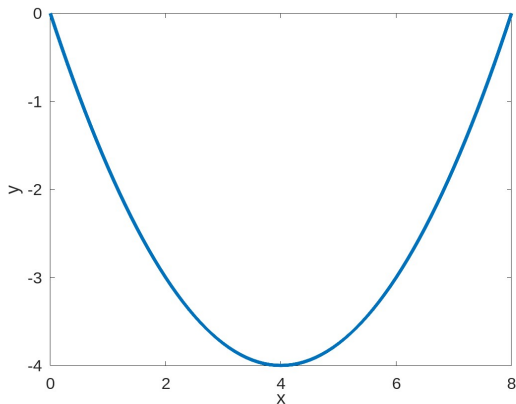
A point-like mobile is moving in the plane (xoy). The Cartesian components of the velocity $V_x(t)$, $V_y(t)$ are given at different times by the table below.

- Let us consider at $y(t = 0) = 0$ $x(0) = 0$ and $y(0) = 0$, calculate x and y .

$t(s)$	0	1	2	3	4
$x(m)$	0	2	4	6	8
$y(m)$	0	-3	-4	-3	0

- Find the mobile trajectory during the time interval $[0, 4]$ (s). Consider: $1 \text{ cm} \rightarrow 1 \text{ m}$.
- Find the acceleration and show on the mobile path, the acceleration vector $\vec{a}$ at $t = 3 \text{ s}$.
- Give the intrinsic component of the acceleration a_t et a_n . Find the radius of the curve ρ .

- The path corresponds to $t \in [0, 4]s$ (scale: $1\text{ cm} \rightarrow 1\text{ m}$)
we plot $y = f(x)$



Acceleration

- The acceleration vector $\vec{a}$ at $t = 3$ s:
 . We have

$$a_x = \frac{dv_x}{dt} \approx \frac{\Delta V_x}{\Delta t} = 0$$

$$a_y = \frac{dv_y}{dt} \approx \frac{\Delta V_y}{\Delta t} = 2 \text{ m/s}^2$$

↑

For the acceleration we have, whatever is the frame

$$\vec{a} = \vec{a}_x + \vec{a}_y = \vec{a}_t + \vec{a}_n$$

but $\vec{a} = \vec{a}_y$ ($a = 2 \text{ m/s}^2$) and :

$$a_t = \frac{dv}{dt} \approx \frac{\Delta v}{\Delta t} = \frac{v(t_2) - v(t_1)}{t_2 - t_1} = \frac{v_2 - v_1}{4 - 2}; \quad (\text{warning}) v = \sqrt{v_x^2 + v_y^2}$$

Intrinsic component

$$a_t = \frac{2\sqrt{5} - 2}{2} = 1.23 \text{ m/s}^2$$

and

$$a_n = \sqrt{a^2 - a_t^2} = \sqrt{4 - 1.23} = 1.57 \text{ /s}^2$$

The radius of the curve is:

$$a_n = \frac{v^2}{\rho} \quad \rho = \frac{v^2}{a} = \frac{16}{1.57} = 10.19 \text{ m}$$